

Braden Lee

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Software Engineer with experience building Python automation, API integrations, cloud-backed applications, and data-driven trading systems. Strong background in reliability troubleshooting for high-availability semiconductor manufacturing environments, with hands-on experience developing dashboards, backend services, authentication tools, and production monitoring systems.

Technical Highlights

- Python automation, backend scripting, and API integration
- Root cause analysis and reliability troubleshooting of complex production systems
- IT help desk support using Active Directory for user/account access administration, end-user troubleshooting, and incident/ticket management
- Assisted with installation and administration of remote monitoring tools and provided support via phone, e-mail, and chat to employees of corporate clients
- Cloud infrastructure experience with AWS EC2, S3, Lambda, CloudWatch, DynamoDB, Cognito; Azure SQL Database, Cosmos DB, Azure Database for MySQL/PostgreSQL; Cloudflare R2 Buckets, Durable Objects, Queues, D1 Database, and KV Namespaces
- Frontend development with JavaScript, React 19, Vite 7, Recharts, and Lucide React icons
- Security and reverse engineering tools including Ghidra, Wireshark, Burp Suite, and Nmap

Skills

- Languages: Python, JavaScript, C++, C, SQL, ASM (x86/x64)
- Frameworks/Libraries: React 19, Vite 7, FastAPI, Django, Recharts, PySide6, PyQtGraph, NumPy, Requests
- Databases/APIs: SQLite, PostgreSQL, TimescaleDB, MongoDB, REST APIs, Robinhood Crypto API
- Tools/Cloud: AWS, Docker, VMware, Ghidra, Wireshark, Nmap, Kali Linux, Tcpdump, GDB

Professional Experience

Intel - Module Equipment Technician (OWLA) | Hillsboro, OR | Dec 2023 - Jul 2025

- Supported reliability and uptime of complex semiconductor manufacturing systems used in wafer level assembly production
- Diagnosed hardware, sensor, and process failures across multi subsystem manufacturing tools using structured troubleshooting and failure isolation
- Executed equipment calibration, verification, and preventive maintenance procedures on production tools, ensuring process control, measurement accuracy, and compliance with manufacturing specifications
- Partnered with process and equipment engineers to perform root cause analysis and implement corrective actions improving long term tool stability
- Achieved L1O certification and Area Coordinator status within two months demonstrating rapid mastery of advanced manufacturing equipment
- Promoted to L2 certification within one year and trained technicians on systematic diagnostics, tool recovery procedures, and reliability practices

Cambia Health Solutions - Software Engineering Intern | Portland, OR | Feb 2021 - Aug 2021

- Developed and maintained Python automation scripts to streamline enterprise document-processing workflows, reduce repetitive manual tasks, and improve operational efficiency
- Implemented, tested, and debugged Agile user-story features while working with internal customers and stakeholders to understand requirements, reported issues, and expected application behavior
- Configured and maintained PsiCapture profiles to automate document capture, structured data extraction, classification, and transfer into enterprise repositories and downstream workflows
- Provided IT and application support for Windows Server environments and enterprise document-processing workflows, troubleshooting software, configuration, access, and system issues
- Worked with Microsoft Azure services and cloud-hosted resources to support application infrastructure, data processing, enterprise integrations, and customer-facing issue resolution
- Utilized SQL databases to query, validate, troubleshoot, and manage application data, communicating technical findings and supporting timely resolution of data and workflow issues

Technical Projects

ASMPT TCB Semiconductor Equipment Maintenance & Troubleshooting

- Provided troubleshooting, repair, preventive maintenance, calibration, testing, and equipment setup for ASMPT thermocompression bonding (TCB) tools
- Diagnosed mechanical, electrical, sensor, wiring, component, and process failures using structured diagnostics, technical documentation, and root cause analysis
- Performed repair, component replacement/installation, adjustment, calibration verification, and post-maintenance functional testing
- Performed scheduled maintenance/inspections to maintain accuracy, repeatability, equipment performance, and production readiness

0x00C0DE Developer Portfolio Website

- Designed and deployed a high-performance static portfolio using pure HTML, CSS, and JavaScript without frameworks
- Engineered a lightweight frontend architecture optimized for speed, responsiveness, and maintainability
- Implemented custom JavaScript components for dynamic UI behavior and interactive content without external dependencies
- Deployed via GitHub Pages with version-controlled CI/CD workflow for rapid updates and reliable uptime

<https://0x00c0de.github.io/>

QR TOTP Authentication System

- Implemented a Python based Time Based One Time Password system compatible with Google Authenticator (RFC 6238)
- Generated QR codes to securely register authentication secrets with authenticator applications
- Used Base32 encoding and HMAC hashing to generate rotating six digit authentication tokens

<https://github.com/0x00C0DE/cs370-proj4-qr-totp>

Collision Avoidance for Autonomous Mobile Robots

- Developed a LiDAR based collision avoidance system using Python
- Utilized OpenCV and Open3D for object and drop off detection
- Generated point clouds from h5py files using time of flight camera data
- Built a visual simulation from generated point clouds to validate detection accuracy

<https://github.com/jwright303/Collision-Avoidance-For-Autonomous-Mobile-Robots>

PROPRTS Robinhood Bot Trading System

- Built an automated stock and options trading platform to collect market data, evaluate multi-interval strategies, manage risk, track portfolio performance, and optionally execute trades through Robinhood
- Implemented Python, robin_stocks, FastAPI, React, Vite, SQLite/PostgreSQL with TimescaleDB support, Docker, Recharts, and optional AI-agent integration
- Developed a real-time dashboard with historical charts, order and transaction ledgers, P&L tracking, PDT protection, configurable controls, manual-trade synchronization, options analysis, and strategy-driven automation

<https://youtu.be/cM1sxCGrlJI>

Robinhood BTC Quantitative Trading Bot

- Developed a Python-based automated cryptocurrency trading and monitoring system integrated with Robinhood's Crypto API for BTC strategy analysis
- Analyzed BTC across multiple intervals using momentum, standard deviation, volume participation, extrapolation, and ensemble signals with configurable risk controls, profit-taking rules, stale-data safeguards, and tracking-only mode
- Built a real-time PySide6/PyQtGraph dashboard for historical charts, limit-order execution, average-cost tracking, P/L visualization, dynamic position sizing, order reconciliation, and configurable safeguards
- Tech stack: Python, NumPy, PySide6, PyQtGraph, REST APIs, Requests, Ed25519 authentication, JSON/JSONL persistence

<https://youtu.be/s9rH4d3edJk>

Education

OREGON STATE UNIVERSITY | B.S. COMPUTER SCIENCE | CORVALLIS, OR | 3.65 GPA | SEPT 2018 - SEPT 2022

- Relevant coursework: Analysis of Algorithms, Data Structures, Operating Systems I and II, Database Systems, Cloud Application Development, Network Security, Computer Architecture and Assembly Language, Cryptography, Defense Against the Dark Arts, Cyber Attack and Defense